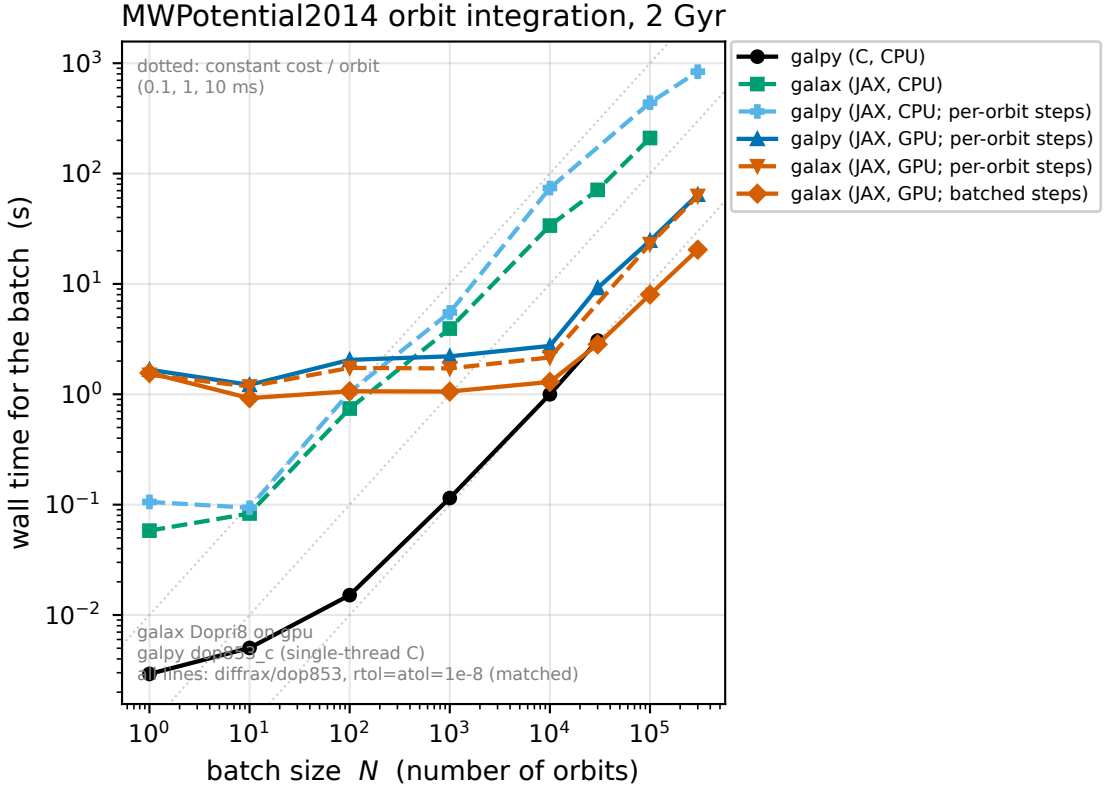
MWPotential2014 orbit integration, 2 Gyr
dotted: constant cost / orbit
(0.1, 1, 10 ms)
galax Dopri8 on gpu
galpy dop853_c (single-thread C)
all lines: diffrax/dop853, rtol=atol=1e-8 (matched)
galpy (C, CPU)
galax (JAX, CPU)
galpy (JAX, CPU; per-orbit steps)
galpy (JAX, GPU; per-orbit steps)
galax (JAX, GPU; per-orbit steps)
galax (JAX, GPU; batched steps)
wall time for the batch (s)
batch size N (number of orbits)
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}